

Honeypot with Threat Intelligence

Honeypot System with Theat Enrichment from Attackers

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Honeyplot with Threat Intelligence

1 Overview

This project is a honeypot system designed to collect information from attackers such as their IP address, HTTP requests and credentials used for brute-force attack. This information is useful for threat intelligence and to know the Tactics, Techniques and Procedures of the attacker. The honeypot system is a server hosting ports 22 and 80. However, no credentials log in to the system.

2 System Architecture

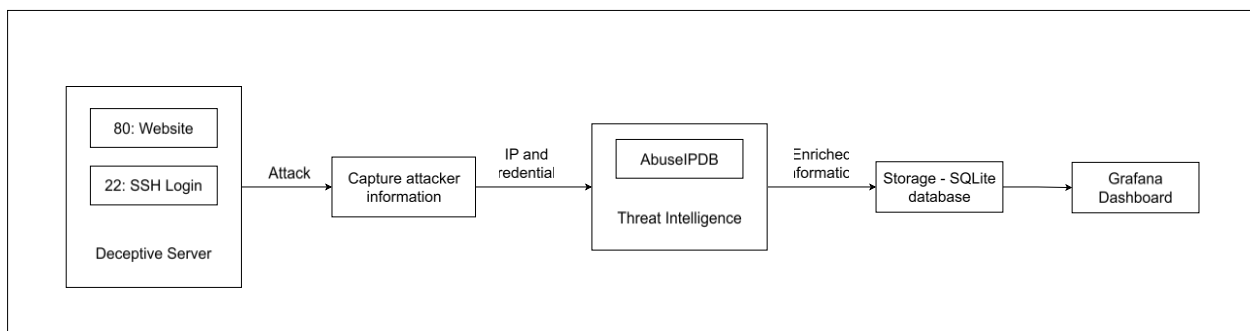


Figure 1 System Architecture

The server hosts ports 22 and 80. When the attacker runs an exploit against these ports, the attacker information is captured. The IP address is queried to AbuseIPDB. The enriched attacker information is stored in SQLite database and the database is fed to Grafana. A dashboard is displayed in Grafana.

3 Technology

1. Python: Python was used to emulate the server and database.
2. Docker: Docker was used to host containerized Grafana.
3. AbuseIPDB APIv2: AbuseIPDB was used to enrich attacker information.
4. Grafana: Grafana was used to visualize the threat intelligence.
5. Kali Linux: Kali Linux was used to attack and test the system.

4 Development

4.1 Honeypot Server

Paramiko is a python library used to implement SSH protocol. It was used to emulate SSH server in server.py program. An RSA server.rsa was generated and used for encryption. http.request was used to intercept the HTTP GET and POST requests on the webpage. Multithreading was used to simultaneously emulate SSH and HTTP connections.

4.2 Threat Intelligence

The IP address of the attacker was retrieved and queried to AbuseIPDB in threat.py. The JSON response was parsed to retrieve the information like confidence score, country, total reports and isp.

4.3 Database

The database.py program initialized the database honeypot.db using SQLite. It provided schema for the attacks table with columnsid, timestamp, ip, attack_type, data_captured, abuse_score, country and isp.

4.4 Dashboard

A dashboard was developed on Grafana consisting of three visualizations. The total attacks was the count of total attacks on the server. The pie chart of Top Targeted Services visualized the services attacked which are SSH and HTTP. The Malicious Ips table visualized the attacker IP addresses with the enriched information from the AbuseIPDB.

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5 Testing

5.1 Setup

The IP address of the machine running the server is 192.168.1.78. Kali Linux was used as an attacker. Its IP address is configured 185.220.101.5.

Kali Linux successfully pinged the server. There was no issue in connectivity.

```
(kali@kali)-[~]
$ ping 192.168.1.78
PING 192.168.1.78 (192.168.1.78) 56(84) bytes of data:
64 bytes from 192.168.1.78: icmp_seq=1 ttl=128 time=1.27 ms
64 bytes from 192.168.1.78: icmp_seq=2 ttl=128 time=1.09 ms
64 bytes from 192.168.1.78: icmp_seq=3 ttl=128 time=1.55 ms
64 bytes from 192.168.1.78: icmp_seq=4 ttl=128 time=0.460 ms
64 bytes from 192.168.1.78: icmp_seq=5 ttl=128 time=0.614 ms
^C
— 192.168.1.78 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4049ms
rtt min/avg/max/mdev = 0.460/0.996/1.549/0.405 ms
(kali@kali)-[~]
$
```

Figure 2 ICMP Ping

5.2 SSH Brute-force

A brute-force attempt was made on port 22 with Hydra via Kali Linux. Following command was used:

```
hydra -l admin -P /usr/share/wordlists/wfuzz/others/common_pass.txt 192.168.1.78 ssh
```

```
(kali@kali)-[~/Desktop]
$ hydra -l admin -P /usr/share/wordlists/wfuzz/others/common_pass.txt 192.168.1.78 ssh
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-17 12:08:43
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 52 login tries (l:l/p:52), ~4 tries per task
[DATA] attacking ssh://192.168.1.78:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-17 12:08:44
```

Figure 3 SSH Brute-force

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Following was the dashboard:

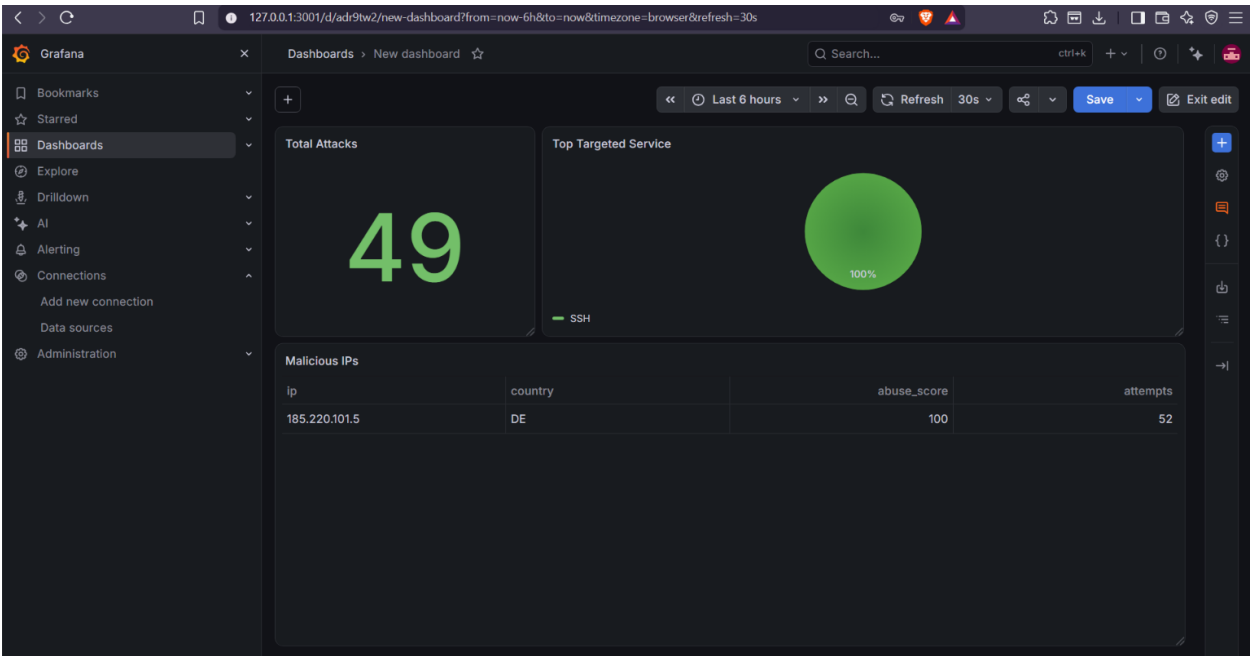


Figure 4 Dashboard for SSH brute-force

Following query was used to see the credentials used for the SSH brute-force:

```
sqlite3 honeypot.db "SELECT ip, data_captured FROM attacks ORDER BY id DESC;"
```

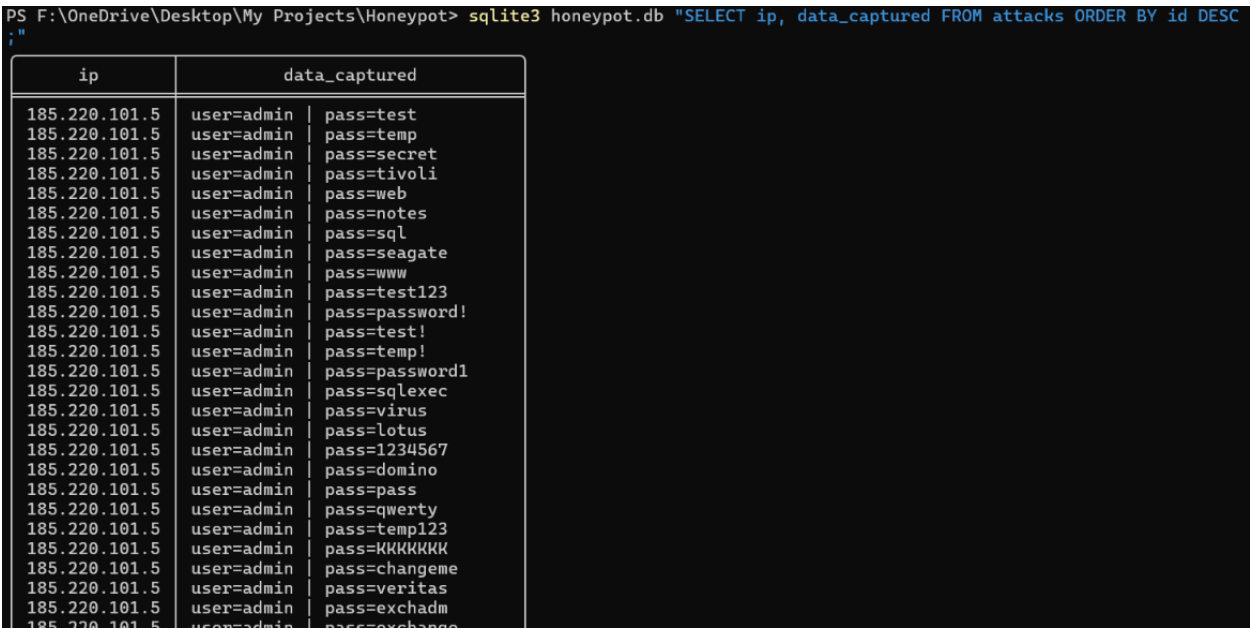


Figure 5 SSH Credentials Attempted

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5.3 Web Scanning

Nikto was used to scan for web vulnerabilities in the server.

```
(kali@kali) [~/Desktop]
$ nikto -h 192.168.1.78
- Nikto v2.6.0

+ Target IP:      192.168.1.78
+ Target Hostname: 192.168.1.78
+ Target Port:    80
+ Platform:      Unknown
+ Start Time:     2026-09-17 12:44:49 (GMT-4)

+ Server: BaseHTTP/0.6 Python/3.11.9
+ ERROR: Failed to check for updates: 403
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.
+ [600652] Python/3.11.9 appears to be outdated (current is at least 3.13.1).
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [000695] /profile.php?u=PD598Rtgr: PowerBoards is vulnerable to path disclosure. See: https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2002-1723
+ [000214] /siteindex/smadmin.html: SiteMinder admin login page available.
+ [000294] /https://classys/users/a/admin/user: SIPS v0.2.2 allows user account info (including password) to be retrieved remotely. See: https://vulners.com/exploitdb/EDB-ID:22381
```

Figure 6 Nikto scan

Following was the dashboard:

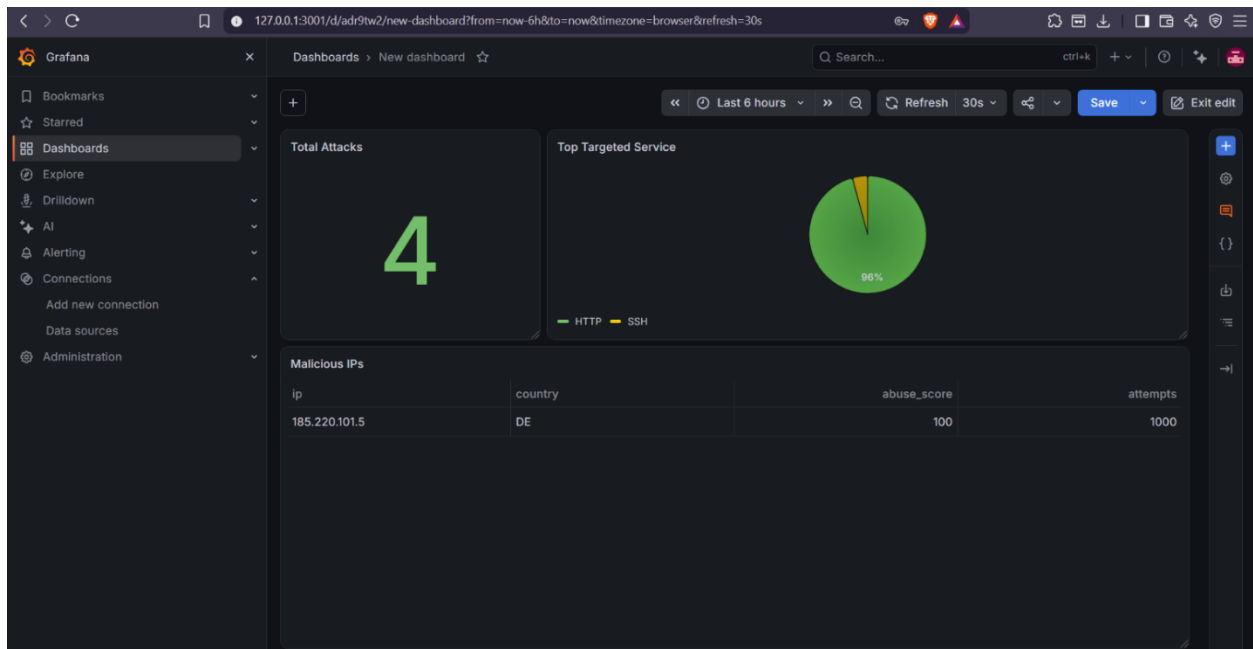


Figure 7 Dashboard for web scanning

Following SQLite query was used to see the 10 requests made by Nikto:

```
sqlite3 honeypot.db "SELECT timestamp, attack_type, data_captured FROM attacks WHERE attack_type = 'HTTP' ORDER BY id DESC LIMIT 10;"
```

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timestamp	attack_type	data_captured
2026-09-17 22:31:46	HTTP	method=GET path=/../../../../../../../../../../../../Windows/win.ini ua=Mozilla/5.0 (iPhone; CPU iPhone OS 18_0 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) EdgiOS/142.0.6281.42 Mobile/15E148 Safari/605.1.15
2026-09-17 22:31:46	HTTP	method=GET path=/cfdocs/exampleapp/docs/sourcewindow.cfm?Template=\\Windows\\win.ini ua=Mozilla/5.0 (iPhone; CPU iPhone OS 18_0 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) CriOS/141.0.6252.91 Mobile/15E148 Safari/604.1
2026-09-17 22:31:46	HTTP	method=GET path=/../../../../winnt/repair/sam_ ua=Mozilla/5.0 (iPhone; CPU iPhone OS 18_0 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/18.0 Mobile/15E148 Safari/604.1
2026-09-17 22:31:46	HTTP	method=GET path=/%2f.%2f.%2f.%2f.%2f.%2f.%2f.%2f.%2f./Windows/win.ini ua=Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:135.0) Gecko/20100101 Firefox/135.0
2026-09-17 22:31:45	HTTP	method=GET path=/ssi/envout.bat? dir%20\\.\\\\.\\\\.\\\\.\\\\.\\\\.\\ ua=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.6282.50 Safari/537.36 OPR/104.0.0.0
2026-09-17 22:31:45	HTTP	method=GET path=/phpping/index.php?pingto=www.example.com%20 %20dir%20c:\\ ua=Mozilla/5.0 (iPhone; CPU iPhone OS 18_0 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) EdgiOS/142.0.6281.42 Mobile/15E148 Safari/605.1.15
2026-09-17 22:31:45	HTTP	method=GET path=/wx/s.dll?d=/Windows/win.ini ua=Mozilla/5.0 (iPhone; CPU iPhone OS 17.5 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/17.5 Mobile/15E148 Safari/604.1

Figure 8 Nikto requests